Text Classification Using K-Nearest Neighbors (K-NN)

Text classification is an essential task in natural language processing (NLP) that involves assigning a category or label to a given piece of text. One straightforward yet effective approach to text classification is the K-Nearest Neighbors (K-NN) algorithm. This method relies on the similarity between texts to determine their classification.

Preprocessing the Text

Before applying K-NN, the text must be preprocessed. This involves:

1. **Cleaning**: Removing unnecessary elements such as punctuation, special characters, and stopwords (common words like "the" and "and").
2. **Tokenizing**: Splitting text into individual words or tokens.
3. **Conversion to Numerical Data**: Since K-NN requires numerical input, the text is transformed using methods like:
   - **Bag-of-Words**: Represents text as a vector of word counts.
   - **TF-IDF (Term Frequency-Inverse Document Frequency)**: A numerical statistic that reflects the importance of a word in a document relative to a collection of documents.

The K-NN Algorithm

K-NN differs from many machine learning algorithms in that it does not have a training phase in the traditional sense. Instead, it involves the following steps:

1. **Storing Labeled Texts**: All texts with known categories are stored in a dataset.
2. **Calculating Distance**: When a new text needs classification, the algorithm computes the distance or similarity between this text and all stored texts. Common distance metrics include Euclidean distance, cosine similarity, and Manhattan distance.
3. **Selecting Nearest Neighbors**: The algorithm identifies the K texts that are nearest (most similar) to the new text.
4. **Assigning a Class**: The new text is assigned to the category that appears most frequently among its K nearest neighbors.

How K-NN Works

In essence, K-NN classifies a new text by examining its proximity to existing texts. The algorithm looks for the most common label among the nearest examples, effectively leveraging the principle of similarity to make its decision.

In summary, K-Nearest Neighbors is a straightforward algorithm that uses the similarity between texts for classification. By leveraging the proximity of labeled examples, it assigns categories based on the most common label among the closest neighbors. This method is particularly useful for those seeking a simple yet effective approach to text classification.